

APPIAH ELLIOT RICHARD

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Interests

Computer / Electronic Engineering, Mathematics & Data Science

Core Competencies

- Python, MATLAB, JavaScript, Vue.js, HTML, CSS
- Data Analysis, Research, UI/UX Design, Figma
- Da Vinci Resolve, Microsoft Suite, Leadership, App
- Inventor, Roblox, Scratch, Minecraft
- Technical Writing, Mentorship, Problem-Solving, Team Collaboration

Education

The Hong Kong University of Science and Technology <i>Electronic Engineering</i>	Aug 2024 – Sep 2028 <i>Hong Kong</i>
Mawuli Senior High School <i>Valedictorian, WASSCE</i>	Sep 2019 – Sep 2022 <i>Ho, Ghana</i>
Mater Ecclesiae School <i>Principal Honor Roll, BECE</i>	Jun 2017 – Sep 2019 <i>Ho, Ghana</i>

Work Experience and Research

Full Stack Developer — HKUST TLE Program <i>Sparrow</i>	Summer 2025 <i>Hong Kong</i>
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- Designed and beautified core pages with Figma, Vue.js, and Vuetify improving visual consistency and mobile accessibility.
- Developed an interactive map search feature with Leaflet.js and Vue2-Leaflet, enabling students to explore 20-30 pre-filled housing listings with filterable categories for enhanced navigation.
- Prototyped comparison page using Vue.js, Axios, CSS Grid.
- Integrated Vue Router and Axios to manage seamless page navigation and data retrieval, ensuring a cohesive and dynamic user experience across the platform.

Video Producer/Content Creator/Writer <i>HKUST Sustainability Office</i>	June 2025 <i>Hong Kong</i>
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- Produced engaging videos/tutorials for sustainability campaigns.
- Used AI tools to create data-driven technical content.

Research Assistant / Program Coordinator <i>Pioneer Academics</i>	Ongoing <i>USA (Remote)</i>
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- S erve as a pivotal liaison between world-class U.S. university professors and diverse international students, delivering 6 hours of weekly mentorship, technical guidance, and coaching to drive innovative research over 12 weeks.s.
- Liaised with professors and mentored students weekly.

Curriculum Developer & Instructor STEMEX Academy, Hong Kong *Ongoing*

- Developed a comprehensive RoboMaster curriculum tailored for high school students in Hong Kong.
- Delivered engaging lessons on RoboMaster, Scratch, and Python to students, fostering technical skills and creativity.
- Prepared students for RoboMaster competitions, guiding them in applying their skills to real-world challenges.

Freelance Coding Instructor <i>Preface School of Technology</i>	Ongoing <i>Hong Kong</i>
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- Spearhead Python, Minecraft, Roblox, Scratch, App Inventor and web development courses, leveraging world class learning materials to deliver personalized, portfolio-ready projects like dynamic web apps and automation tools.
- Mentored diverse learners and led workshops.

R&D Assistant <i>Voyager Education</i>	Feb 2024 <i>Hong Kong</i>
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- Developed robot/software modules and STEAM programs.

Web Developer/Programmer

Pet Pet Club Ltd.

- Revamped website and evaluated deep learning models.

Dec 2024

Hong Kong

Student Ambassador

HKUST

- Led campus tours, supported events and outreach.

Ongoing

Hong Kong

Intern

Daystar Power Solutions

- Installed 4.3MW plant, saving 30% in energy/emissions.

Oct 2023

Kumasi, Ghana

Researcher

KNUST with Dr. Adu Boakye

- Built ambulance queuing model now used by 5 hospitals.

Jul 2023

Kumasi, Ghana

Awards and Honors

- HKSAR Belt and Road Scholarship, HKUST Global Learner's Award
- Innovation Excellence Award, STEM Hero, NSMQ Valedictorian
- Ghana Science Olympiad Gold Medalist, Tech-Up Africa Distinction
- Dr. Adu Boakye Math/Physics/Chemistry Award

Leadership Experience

Founder

Elliot Explains

- Created animated science videos (170k+ views), supported 5 NSMQ teams.

2020

YouTube

Team Captain

National Science & Maths Quiz

- Led team to Top 9 nationally; secured learning center renovations.

2022

Ghana

President

National Sharks Quiz

- Grew club 5x, raised funds, planned weekly drills.

2022

Ghana

Co-Founder & VP

IT Hub

- Revamped website, built E-voting system.

2021

Ghana

Operations Manager

Kubi Foundation

- Managed clinic/borehole projects, empowered girls with tie-dye skills.

2022

Ghana

Founder

Kindred Hearts

- Led beach cleanup, recycling, and disaster donations.

2023

Ghana

Tutor

Aikins Educational Consult

- Tutored SAT/ACT Math and Science.

Aug 2023

Accra, Ghana

Academic Competitions

- NSMQ Regional Champion & Semifinalist, National Sharks Quiz Runner-up
- UCMAS Regional Champion, Ghana Science Olympiad Gold Medalist
- UG Numeric Nexus Winner, National Code Fest Winner

Personal Attributes

- High analytical proficiency, adept in data-driven projects.
- Calm under pressure with strong communication and leadership.
- Strong interpersonal and presentation skills.